Reg. No. | | | | | | | | | | | [12EE74]

# SEVENTH SEMESTER B.E EXAMINATION, DECEMBER-2015

## ELECTRICAL DRIVES AND APPLICATIONS

**Time:3 Hours** **Maximum Marks:70**

**Instructions:** Answer any **FIVE** full Questions.

1. a. What are the advantages of an Electrical Drive? **06**

   b. With a neat diagram, explain the speed torque conventions and multi quadrant operations of a motor, driving a hoist load. **08**

2. a. A motor operates on a periodic duty cycle in which it is clutched to its load for 10 minutes and declutched to run on load for 20 minutes. Minimum temperature rise is 40° C. Heating and cooling time constants are equal and have a value of 60 minutes. When load is declutched continuously the temperature rise is 15° C. Determine: **06**

   i. Maximum temperature during the duty cycle, and

   ii. Temperature when the load is clutched continuously.

   b. Explain the standard classes of motor duties with load diagrams. **08**

3. a. With a neat circuit and graph, explain the regenerative and plugging type of braking system for separately excited DC shunt motor. **07**

   b. A 220 V, 1500 rpm, 50 A separately excited motor with armature resistance of 0.52 Ω, is fed from a 3-phase fully-controlled rectifier. Available ac source has line voltage of 440 V, 50 Hz. A star-delta connected transformer is used to feed the armature so that motor terminal voltage equal rated voltage when converter firing angle is zero. **07**

   i. Calculate transformer turns ratio

   ii. Determine he value of firing angle when motor is running at 1200 rpm and rated torque

4. a. Explain the effect of unbalanced source voltages and single phasing on three phase Induction motor. **08**

   b. Explain the regenerative braking of an induction motor. **06**

5. a. Explain the operation of voltage source inverter fed induction motor drive with neat circuit and wave forms. **07**

   b. With a neat driver circuit, explain the static Scherebius drive. **07**

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6. a. With a neat block diagram, explain the true synchronous mode variable frequency control of multiple synchronous motor. **07**

b. A synchronous motor is controlled by a load commutated converter. Source voltage is 6.6 KV, 50 Hz. Load commutated inverter operates at a constant firing angle $\alpha_l$ 140° and when rectifying $\alpha_l = 0°$. Dc link inductor resistance $R_d = 0.1\Omega$. Drive operates in self-control mode with a constant (v/f) ratio. Motor has the details: 8 MW, 3-phase, 6600 V, 6 pole, 50 Hz, unity power factor, star connected, $x_s = 2.8\Omega$, $R_s = 0$. Determine source side converter firing angle for motor operation at the rated current and 500 rpm. What will be the power developed by motor? **07**

7. a. With schematic, explain the wire section in a paper mill drive. **07**

b. Explain the driving motors used in the cement industry for different operations. **07**

8. a. Mention the losses in electrical drive system. **06**

b. What are the measures for energy conservation in electrical drives? **08**